5. XGBoost

The classifier uses the three columns chosen in the previous chapter: VolumeLag1, Ret1, CloseEMA8. Optuna runs 100 trials on the 3,723 development sessions. Each trial is a different combination of tree depth, minimum child weight, row subsample, learning rate, and number of trees. The next trial is aimed near combinations that already scored well. The score is the mean ROC AUC across 5 time-series folds, with a gap of 10 sessions. The subsample is kept between 0.5 and 1. The seed is 21. The hold-out file is not read.

The best mean AUC on the development folds is 0.5335, with a spread across folds of 0.0307. The chosen settings are depth 9, minimum child weight 9, subsample 0.595, learning rate 0.013, and 925 trees.

Ten best trials

Rank	Mean AUC	Depth	Min child	Subsample	Learning rate	Trees
1	0.5335	9	9	0.595	0.013	925
2	0.5334	8	9	0.576	0.017	881
3	0.5323	9	9	0.573	0.034	676
4	0.5321	8	9	0.555	0.023	845
5	0.5318	9	9	0.513	0.018	889
6	0.5318	9	9	0.592	0.012	991
7	0.5317	9	9	0.548	0.034	753
8	0.5315	9	9	0.573	0.014	996
9	0.5312	8	8	0.524	0.016	847
10	0.5310	9	8	0.555	0.025	791